The pairwise-deletion weight matrix for WLSMV with ordinal missing data

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Purpose

WLSMV is limited information: its sample statistics are the univariate thresholds and bivariate polychoric correlations, so it can absorb missing data by computing each statistic from its available cases (pairwise deletion, PD). The point estimates are then standard; the open question is the weight matrix. What is the asymptotic covariance Γ^{pw} of the PD statistic vector, and how does it differ from the complete-data Γ that the conventional "WLSMV with PD" plugs in?

Answer. Γ^{pw} is the complete-data Γ reweighted by observation-overlap proportions. Writing π_a for the probability that a case observes every item needed for statistic a and π_{ab} for the probability it observes the items needed for both a and b,

$$\Gamma^{\mathrm{pw}}_{ab} \;=\; \frac{1}{\pi_a \pi_b}\, \mathbb{E}\Big[\mathbf{1}\{S_a \cup S_b \subseteq R\}\, \psi^a \psi^b\Big], \qquad \text{(Theorem 1)}$$

which under MCAR collapses to the Hadamard form $\Gamma^{\mathrm{pw}}_{ab} = (\pi_{ab}/\pi_a\pi_b)\,\Gamma_{ab}$, with diagonal $\Gamma^{\mathrm{pw}}_{aa} = \Gamma_{aa}/\pi_a$. This is the ordinal sibling of the continuous covariance rule $\Gamma^{\mu,\mathrm{pw}}_{jk} = \sigma_{jk}\,\pi_{jk}/\pi_j\pi_k$ already used on the pairwise-GLS path. The conventional implementation omits the reweighting (it uses one nominal N), so it understates the variance of statistics estimated from fewer cases; that is the documented source of the inflated $\Delta\chi^2$ [6].

Setup and notation

Let there be p ordinal items, item j with C_j categories and thresholds $\tau_j \in \mathbb{R}^{C_j-1}$. The WLSMV statistic vector stacks all thresholds and all polychorics,

$$s = (\tau', \rho')', \qquad \rho = (\rho_{jk})_{j<k}, \qquad p^* = \sum_j (C_j - 1) + \binom{p}{2}.$$

Index a generic component by $a \in \{1, \dots, p^*\}$ and let $S_a \subseteq \{1, \dots, p\}$ be its *support*: $|S_a| = 1$ for a threshold of item j ($S_a = \{j\}$), $|S_a| = 2$ for a polychoric ($S_a = \{j, k\}$).

Complete data are i.i.d. $z_i = (y_{i1}, \dots, y_{ip})$. The staged estimator (univariate thresholds, then bivariate polychorics with thresholds held fixed [1, 2]) is an M-estimator solving $\sum_i g(z_i; \hat{s}) = 0$, with bread $A = -\mathbb{E}[\partial g/\partial s']$ and casewise influence

$$\psi_i = A^{-1} g(z_i; s_0), \qquad \mathbb{E}[\psi_i] = 0, \qquad \Gamma = \mathbb{E}[\psi_i \psi_i'].$$

magmaan stores the influence rows $IF_i = -\psi_i A^{-\top}$ and forms $\widehat{\Gamma} = \frac{1}{N} \sum_i IF_i IF_i'$ (cf. `robust_ordinal_gamma.tex`).

Missingness is encoded by the response set $R_i = \{j : y_{ij} \text{ observed}\}$. Statistic a is estimable on case i iff $S_a \subseteq R_i$; write the availability indicator and its proportions

$$d_i^a = \mathbf{1}\{S_a \subseteq R_i\}, \quad \pi_a = \mathbb{E}[d_i^a], \quad \pi_{ab} = \mathbb{E}[d_i^a d_i^b] = \Pr(S_a \cup S_b \subseteq R), \quad N_a = \sum_i d_i^a.$$

PD solves each block on its available cases; stacking with the diagonal mask $D_i = \operatorname{diag}(d_i^a)$, the estimator solves $\sum_i D_i\, g(z_i; \hat{s}^{\text{pw}}) = 0$.

Influence sparsity

Lemma 1. *The influence ψ_i^a depends on z_i only through $\{y_{ij} : j \in S_a\}$. Hence $\psi_i^{\tau_j}$ is a function of y_{ij} alone, and $\psi_i^{\rho_{jk}}$ a function of (y_{ij}, y_{ik}) alone.*

Proof. The stacked score is block lower triangular: threshold scores depend on their own item; the ρ_{jk} score depends on the pair (j, k) and on τ_j, τ_k. Thus A is block lower triangular and so is A^{-1}, giving

$$\psi_i^{\rho_{jk}} = \frac{1}{\mathcal{I}_{\rho_{jk}}} \Big[s_{\rho_{jk}}(y_{ij}, y_{ik}) - \mathcal{I}_{\rho_{jk}\tau_j} \psi_i^{\tau_j} - \mathcal{I}_{\rho_{jk}\tau_k} \psi_i^{\tau_k} \Big],$$

a combination of objects supported on $\{j, k\}$; the threshold rows are supported on their single item. $\square$

Lemma 1 is what makes overlap proportions the right bookkeeping: the only cases that carry information about the pair (a, b) are those that observe $S_a \cup S_b$.

The pairwise weight matrix

Theorem 1. *Let* $s_0 = \operatorname{plim} \hat{s}^{\mathrm{pw}}$ *and let* ψ^a *be the influence about* s_0*. Then* $\sqrt{N}(\hat{s}^{\mathrm{pw}} - s_0) \xrightarrow{d} N(0, \Gamma^{\mathrm{pw}})$ *with*

$$\Gamma^{\mathrm{pw}}_{ab} = \frac{1}{\pi_a \pi_b} \mathbb{E}\Big[\mathbf{1}\{S_a \cup S_b \subseteq R\}\, \psi^a \psi^b\Big].$$

Proof. The available-case linearization of component a is $\hat{s}^{\mathrm{pw}}_a - s_{a,0} = N_a^{-1} \sum_i d_i^a \psi_i^a + o_p(N^{-1/2})$; the mask is mean-zero against the influence at the pseudo-true value, $\mathbb{E}[d_i^a \psi_i^a] = A^{-1}\mathbb{E}[d_i^a g_a(z_i; s_0)] = 0$, by definition of s_0. Independent cross terms vanish, and with $N_a = N\pi_a + o_p(N)$,

$$N\ \mathrm{Cov}(\hat{s}^{\mathrm{pw}}_a, \hat{s}^{\mathrm{pw}}_b) = \frac{N}{N_a N_b} \sum_i d_i^a d_i^b \psi_i^a \psi_i^b + o_p(1) \xrightarrow{p} \frac{1}{\pi_a \pi_b} \mathbb{E}[d^a d^b \psi^a \psi^b],$$

and $d_i^a d_i^b = \mathbf{1}\{S_a \cup S_b \subseteq R_i\}$. □

Corollary 1 (MCAR closed form)**.** *If* $R \perp z$ *then* s_0 *is the true value and the indicator factors out:*

$$\Gamma^{\mathrm{pw}}_{ab} = \frac{\pi_{ab}}{\pi_a \pi_b} \Gamma_{ab}, \qquad \textit{equivalently} \qquad \Gamma^{\mathrm{pw}} = \Pi_{\div} \odot \Gamma, \quad (\Pi_{\div})_{ab} = \frac{\pi_{ab}}{\pi_a \pi_b},$$

with diagonal $\Gamma^{\mathrm{pw}}_{aa} = \Gamma_{aa}/\pi_a$*, i.e.* $\mathrm{Var}(\hat{s}^{\mathrm{pw}}_a) = \Gamma_{aa}/N_a$*.*

Corollary 2 (item-independent missingness)**.** *If items are observed independently with* $\Pr(j \in R) = \pi_j$*, then* $\pi_a = \prod_{j \in S_a} \pi_j$ *and the overlap factor is the product of the inverse observation probabilities of the* shared *items,*

$$\frac{\pi_{ab}}{\pi_a \pi_b} = \prod_{j \in S_a \cap S_b} \pi_j^{-1}.$$

In particular the factor is 1 *for statistics with disjoint supports, and the correction touches only diagonals and entries sharing an item.*

The diagonal in Corollary 1 is the whole story for Type I error: a statistic seen by $N_a < N$ cases has variance Γ_{aa}/N_a, not Γ_{aa}/N. Replacing N_a by N understates that variance, overstates the corresponding precision in the weight, and inflates the quadratic-form statistic. This is exactly the WLSMV_PD behaviour reported in [6] (Type I rate of $\Delta\chi^2$ up to 0.55 at 50% missing).

Remark (the correction preserves positive semidefiniteness). $\Pi = [\pi_{ab}]$ is a Gram matrix ($\pi_{ab} = \mathbb{E}[d^a d^b]$), hence PSD; conjugating by $\mathrm{diag}(\pi_a^{-1})$ keeps it PSD, so $\Pi_{\div}$ is PSD, and by the Schur product theorem $\Gamma^{\mathrm{pw}} = \Pi_{\div} \odot \Gamma$ is PSD. The overlap reweighting cannot by itself create an indefinite weight; any non-PSD-ness is inherited from Γ and handled by the existing repair policy (`robust_ordinal_gamma.tex`).

Remark (exact staged sandwich vs. the Hadamard form). Theorem 1 accounts each statistic at the level of its own influence, which assigns the entire polychoric influence the pair availability π_{jk}. The fully exact object is the masked sandwich

$$\Gamma^{\mathrm{pw}} = \bar{A}^{-1} M \bar{A}^{-\top}, \quad \bar{A} = \mathbb{E}[D_i G_i], \quad M_{ab} = \mathbb{E}[\mathbf{1}\{S_a \cup S_b \subseteq R\}\, g_a g_b'], \quad G_i = \frac{\partial g}{\partial s'},$$

which assigns the threshold-propagation terms of $\psi^{\rho_{jk}}$ (Lemma 1) the larger item availabilities $\pi_j, \pi_k \geq \pi_{jk}$. The Hadamard form is therefore exact on the threshold block and a mild upper bound on the polychoric block; the gap is of order $\mathcal{I}_{\rho\tau}\,(\pi_{jk}^{-1} - \pi_j^{-1})$ and vanishes when thresholds are well determined. Because it overstates the polychoric variances, the Hadamard estimator errs *conservative* (it under-rejects), the safe direction for a Type I criterion, and it is the cheap option built directly on the existing IF rows.

Remark (scope: this calibrates inference, not PD's MAR bias). Γ^{pw} is the sampling covariance of the PD statistic about its own probability limit s_0. Available-case polychorics are consistent for the true s under MCAR but not generally under MAR, where selection on an observed, latent-correlated item shifts s_0 off the target. The corrected weight then yields a correctly calibrated test of a slightly displaced pseudo-value; it does not restore MAR consistency, which needs full information (FIML / two-stage) or the composite-likelihood machinery with an MAR-valid sandwich [5]. Empirically the displacement was small in [6] (loading bias $|\mathrm{RB}| < 0.1$), so the inference failure, not the point estimate, dominated; the corrected Γ^{pw} targets exactly that failure.

What the conventional implementation omits

Plugging the PD point estimates into the complete-data Γ formula at a single sample size is, in the language of Corollary 1, the choice $\pi_a \equiv \pi_b \equiv \pi_{ab}$, i.e. $\Pi_{\div} \equiv \mathbf{1}\mathbf{1}'$ and $\Gamma^{\mathrm{pw}} \equiv \Gamma$. The non-uniform N_a across thresholds (item level) and polychorics (pair level), and the missing-data covariance between them, are discarded. Two consequences follow, both noted in [6]: the per-statistic variances are wrong (the diagonal effect above), and the correlations among statistics estimated on different overlapping subsamples are wrong (the off-diagonal effect). Neither the mean nor the mean-and-variance χ^2 correction can compensate, because both read their scaling from this same matrix.

Estimation

The leading-order estimator is the existing influence machinery with overlap masks. For statistics a, b let $\widehat{\psi}_i^a$ be the centered PD influence rows; then

$$\widehat{\Gamma}^{\mathrm{pw}}_{ab} = \frac{N}{N_a N_b} \sum_{i:\, S_a \cup S_b \subseteq R_i} \widehat{\psi}_i^a\, \widehat{\psi}_i^b.$$

This is $\widehat{\Gamma} = IF'IF/N$ with the sum restricted to the overlap set and the $1/N$ replaced by $N/N_a N_b$; under MCAR it converges to Corollary 1, under MAR to Theorem 1. Implementation is pattern-grouped, matching the continuous pairwise path: partition cases by distinct R_i, accumulate $\widehat{\psi}^a \widehat{\psi}^b$ over patterns whose set contains $S_a \cup S_b$, and divide once at the end. The two-level structure is automatic from the supports: $\{j\}$ for thresholds feeds the item proportions π_j, $\{j, k\}$ for polychorics feeds the pair proportions π_{jk}, and the mixed block uses $\pi_{\{j,k,l\}}$. The exact sandwich (remark above) replaces the own-influence rows by the masked score-and-bread accumulation $\bar{A}^{-1} M \bar{A}^{-\top}$ when the conservative bias is judged too large.

Downstream: test statistic and FMG spectrum

Everything past the weight matrix is the single substitution $\Gamma \mapsto \Gamma^{\mathrm{pw}}$. Fit DWLS with weight $V = W_D^{-1}$, $W_D = \mathrm{diag}(\Gamma^{\mathrm{pw}})$; the residual $\hat{e} = \hat{s} - \sigma(\hat{\theta})$ gives $T = N\, \hat{e}' V \hat{e}$, and

$$T \xrightarrow{d} \sum_{a=1}^{df} \lambda_a\, \chi^2_{1,a}, \qquad \{\lambda_a\} = \mathrm{eig}(U\Gamma^{\mathrm{pw}}), \qquad U = V - V\Delta(\Delta' V \Delta)^{-1} \Delta' V,$$

with $\Delta = \partial\sigma/\partial\theta$. The mean correction (WLSM) is $\hat{c} = \mathrm{tr}(U\Gamma^{\mathrm{pw}})/df$; the mean-and-variance correction (WLSMV) matches $\mathrm{tr}(U\Gamma^{\mathrm{pw}})$ and $\mathrm{tr}((U\Gamma^{\mathrm{pw}})^2)$ [3]; and the FMG / pEBA test uses the full penalized spectrum of $U\Gamma^{\mathrm{pw}}$ rather than its first two moments [7]. The nested $\Delta\chi^2$ for measurement invariance inherits the same substitution. The default's error is precisely $U\Gamma$ in place of $U\Gamma^{\mathrm{pw}}$; with the corrected meat the WLSMV missing-data statistic is the analogue of Savalei and Bentler's statistically justified pairwise method for continuous nonnormal incomplete data [4], and an overlap-weighted Γ^{pw} is exactly what an ordinal FMG difference test under missingness would consume.

References

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